

for all primes p , then $A = 0$. For the short exact sequences $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{Z}_p \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$, the six-term exact sequences in (6) of the proposition become

$$\begin{aligned} 0 \rightarrow \operatorname{Tor}(A, \mathbb{Z}_p) \rightarrow A \xrightarrow{p} A \rightarrow A \otimes \mathbb{Z}_p \rightarrow 0 \\ 0 \rightarrow \operatorname{Tor}(A, \mathbb{Q}/\mathbb{Z}) \rightarrow A \rightarrow A \otimes \mathbb{Q} \rightarrow A \otimes \mathbb{Q}/\mathbb{Z} \rightarrow 0 \end{aligned}$$

If $\operatorname{Tor}(A, \mathbb{Z}_p) = 0$ for all p , then exactness of the first sequence implies that $A \xrightarrow{p} A$ is injective for all p , so A is torsionfree. Then $\operatorname{Tor}(A, \mathbb{Q}/\mathbb{Z}) = 0$ by (3) or (4) of the proposition, so the second sequence implies that $A \rightarrow A \otimes \mathbb{Q}$ is injective, hence $A = 0$ if $A \otimes \mathbb{Q} = 0$. $\square$

The algebra by means of which the Tor functor is derived from tensor products has a very natural generalization in which abelian groups are replaced by modules over a fixed ring R with identity, using the definition of tensor product of R -modules given in §3.2. Free resolutions of R -modules are defined in the same way as for abelian groups, using free R -modules, which are direct sums of copies of R . Lemmas 3A.1 and 3A.2 carry over to this context without change, and so one has functors $\operatorname{Tor}_n^R(A, B)$. However, it need not be true that $\operatorname{Tor}_n^R(A, B) = 0$ for $n > 1$. The reason this was true when $R = \mathbb{Z}$ was that subgroups of free groups are free, but submodules of free R -modules need not be free in general. If R is a principal ideal domain, submodules of free R -modules are free, so in this case the rest of the algebra, in particular the universal coefficient theorem, goes through without change. When R is a field F , every module is free and $\operatorname{Tor}_n^F(A, B) = 0$ for $n > 0$ via the free resolution $0 \rightarrow A \rightarrow A \rightarrow 0$. Thus $H_n(C \otimes_F G) \approx H_n(C) \otimes_F G$ if F is a field.

Exercises

1. Use the universal coefficient theorem to show that if $H_*(X; \mathbb{Z})$ is finitely generated, so the Euler characteristic $\chi(X) = \sum_n (-1)^n \operatorname{rank} H_n(X; \mathbb{Z})$ is defined, then for any coefficient field F we have $\chi(X) = \sum_n (-1)^n \dim H_n(X; F)$.
2. Show that $\operatorname{Tor}(A, \mathbb{Q}/\mathbb{Z})$ is isomorphic to the torsion subgroup of A . Deduce that A is torsionfree iff $\operatorname{Tor}(A, B) = 0$ for all B .
3. Show that if $\tilde{H}^n(X; \mathbb{Q})$ and $\tilde{H}^n(X; \mathbb{Z}_p)$ are zero for all n and all primes p , then $\tilde{H}_n(X; \mathbb{Z}) = 0$ for all n , and hence $\tilde{H}^n(X; G) = 0$ for all G and n .
4. Show that $\otimes$ and Tor commute with direct limits: $(\varinjlim A_\alpha) \otimes B = \varinjlim (A_\alpha \otimes B)$ and $\operatorname{Tor}(\varinjlim A_\alpha, B) = \varinjlim \operatorname{Tor}(A_\alpha, B)$.
5. From the fact that $\operatorname{Tor}(A, B) = 0$ if A is free, deduce that $\operatorname{Tor}(A, B) = 0$ if A is torsionfree by applying the previous problem to the directed system of finitely generated subgroups A_α of A .
6. Show that $\operatorname{Tor}(A, B)$ is always a torsion group, and that $\operatorname{Tor}(A, B)$ contains an element of order n iff both A and B contain elements of order n .

3.B The General Künneth Formula

Künneth formulas describe the homology or cohomology of a product space in terms of the homology or cohomology of the factors. In nice cases these formulas take the form $H_*(X \times Y; R) \approx H_*(X; R) \otimes H_*(Y; R)$ or $H^*(X \times Y; R) \approx H^*(X; R) \otimes H^*(Y; R)$ for a coefficient ring R . For the case of cohomology, such a formula was given in Theorem 3.16, with hypotheses of finite generation and freeness on the cohomology of one factor. To obtain a completely general formula without these hypotheses it turns out that homology is more natural than cohomology, and the main aim in this section is to derive the general Künneth formula for homology. The new feature of the general case is that an extra Tor term is needed to describe the full homology of a product.

The Cross Product in Homology

A major component of the Künneth formula is a **cross product** map

$$H_i(X; R) \times H_j(Y; R) \xrightarrow{\times} H_{i+j}(X \times Y; R)$$

There are two ways to define this. One is a direct definition for singular homology, involving explicit simplicial formulas. More enlightening, however, is the definition in terms of cellular homology. This necessitates assuming X and Y are CW complexes, but this hypothesis can later be removed by the technique of CW approximation in §4.1. We shall focus therefore on the cellular definition, leaving the simplicial definition to later in this section for those who are curious to see how it goes.

The key ingredient in the definition of the cellular cross product will be the fact that the cellular boundary map satisfies $d(e^i \times e^j) = de^i \times e^j + (-1)^i e^i \times de^j$. Implicit in the right side of this formula is the convention of treating the symbol $\times$ as a bilinear operation on cellular chains. With this convention we can then say more generally that $d(a \times b) = da \times b + (-1)^i a \times db$ whenever a is a cellular i -chain and b is a cellular j -chain. From this formula it is obvious that the cross product of two cycles is a cycle. Also, the product of a boundary and a cycle is a boundary since $da \times b = d(a \times b)$ if $db = 0$, and similarly $a \times db = (-1)^i d(a \times b)$ if $da = 0$. Hence there is an induced bilinear map $H_i(X; R) \times H_j(Y; R) \rightarrow H_{i+j}(X \times Y; R)$, which is by definition the cross product in cellular homology. Since it is bilinear, it could also be viewed as a homomorphism $H_i(X; R) \otimes_R H_j(Y; R) \rightarrow H_{i+j}(X \times Y; R)$. In either form, this cross product turns out to be independent of the cell structures on X and Y .

Our task then is to express the boundary maps in the cellular chain complex $C_*(X \times Y)$ for $X \times Y$ in terms of the boundary maps in the cellular chain complexes $C_*(X)$ and $C_*(Y)$. For simplicity we consider homology with $\mathbb{Z}$ coefficients here, but the same formula for arbitrary coefficients follows immediately from this special case. With $\mathbb{Z}$ coefficients, the cellular chain group $C_i(X)$ is free with basis the i -cells of X , but there is a sign ambiguity for the basis element corresponding to each cell e^i ,